

Memory Forensics with Volatility

Why Memory Knows Everything

RAM contains the live truth:

Running processes (including "fileless" malware that never touched disk)

Loaded modules / DLLs

Open network connections

Command-line arguments (the attacker's actual commands)

Crypto keys / passwords cached by tools

Volatility 3 Quick Start

```
# vol3 is plugin-based; no profile needed
vol3 -f mem.img windows.pslist.PsList
vol3 -f mem.img windows.psscan.PsScan          # deeper scan of process objects
vol3 -f mem.img windows.cmdline.CmdLine        # the attacker's command lines!
vol3 -f mem.img windows.netstat.NetStat        # connections
vol3 -f mem.img windows.malfind.Malfind        # injected code
vol3 -f mem.img windows.modscan.ModScan        # hidden modules
```

The "Process Tree" Read

Print the tree (pslist): browsers under word.exe? An "unusual parent-child pair" is the classic tell.

winword.exe → powershell.exe = macro infection!

svchost.exe → whoami /net user = someone poking inside

rundll32.exe with a random .dll = suspicious

```
vol3 -f mem.img windows.pstree.PsTree
```

cmdline: The Anciest Jewel

The command lines in memory tell the story OUTRIGHT:

```
"powershell -enc DQBJAGwAYQBzAHMA..." <- base64 payload  
"reg add HKLM\\...\\Run /v updater"
```

Decode that base64 and you've pretty much solved it.

netstat: Contact the C2

```
vol3 -f mem.img windows.netstat.NetStat | grep ESTABLISH  
# outbound connection to 185.x.x.x on 443 from powershell.exe = C2 hopper
```

malfind: The Hidden Injection

Memory-only malware injects into legit processes. **Malfind** flags regions with executable permissions + suspicious content (Read/Write/Execute = rwx).

```
vol3 -f mem.img windows.malfind.Malfind
```

Then dump the suspicious region:

```
vol3 -f mem.img windows.dumpfiles.DumpFiles -p <pid>
```

From Dump to Verdict

Extract the injected blob

Scrape strings / hash it

Look it up (VT) or disassemble – now you have the "what".

The DFIR arrow: raw memory → pslist/cmdline → malfunction → blob → identify. That's the whole day, done right.